BOAMP Renewal Linkage — Layer 2 (`enriched`, formerly M1)

Buyer-SIREN Enrichment, Alias Bridge, and Controlled Comparison with Layer 1

Gigalis Predictive-Modeling Internship — Current Independent Study

Repository state of July 2026; M1 outputs generated 2026-07-15

Revision note (v2, 2026-07-17 — two-layer restructure). This report was regenerated after the repository was restructured into two controlled layers: **Layer 1 =** `boamp_only` **(formerly M0)** and **Layer 2 =** `enriched` **(formerly M1)**. The full pipeline was rerun end-to-end on 2026-07-17 from `notebooks/01_data_engineering_pipeline.ipynb` (logic extracted, parity-verified, into `src/boamp/`); every headline number in this report was reproduced exactly: 84,623 cleaned notices, 3,159 eligible sources, 6,137 Layer-1 candidate pairs, balanced links 618 (19.6%) / Layer-2 847 (26.8%), thresholds 0.2642/0.3230/0.3931. Corrections established during the rerun: (i) the cleaned corpus has **84,623** logical rows — the 87,418 figure in the v1 reproducibility manifest was a raw line-count artifact caused by embedded newlines in quoted text fields (same artifact behind the "3,170-row" M1 table entry; the true count is 3,159); (ii) the raw input is **139** monthly data files plus a metadata JSON (the "140" in v1 audit tables counted directory entries); (iii) the previously empty `survival_logrank_tests.csv` is now populated (the v1 script compared float-typed CPV divisions against string literals); (iv) links now carry an explicit three-way confidence tier — 38% of Layer-1 and 39% of Layer-2 balanced links are POTENTIAL (top1–top2 margin < 0.05). Legacy file paths cited in the body (`boamp_m0_*`, `boamp_m1_*`, `m0_*/m1_*` tables) now live under `archive/` (see `archive/ARCHIVE_MANIFEST.csv`); their two-layer successors are under `data/processed/{boamp_only, enriched, comparison}/`.

Abstract

This report is the consolidated technical reference for the M1 buyer-identity enrichment branch. M1 is not the official baseline. The official study result remains M0, the BOAMP-only balanced specification. M1 is a sensitivity analysis designed to answer one controlled question: *how much does buyer-name fragmentation in M0 affect candidate coverage and proxy-recurrence counts when buyer identity is enriched with validated SIREN evidence?*

M1 keeps the M0 eligible source population, temporal window logic, TF–IDF object-text score, CPV score, temporal score, ranking rule, and balanced threshold. The controlled change is buyer reconciliation. M1 uses exact BOAMP SIRET when present, then a validated SIREN from either a direct notice-level enrichment join or a conservative historical name–department alias bridge, and finally the original M0 normalized-name fallback. The M0 balanced threshold is reused unchanged (0.323022), so M1's higher event count is not produced by relaxing the decision threshold.

On the same 3,159 eligible digital/ICT source notices, M1 increases blocking coverage from 39.1% to 47.6%, candidate pairs from 6,137 to 8,228, and accepted balanced proxy-recurrence links from 618 to 847. The overall proxy-recurrence rate rises from 19.6% to 26.8%. This is useful evidence that buyer identifiers matter, but it is not proof that the incremental links are legal renewals. M1 adds 273 links not present in M0, loses 44 M0 links under the new ranking, and has link-set Jaccard overlap 0.644 with M0. Incremental links have weaker text evidence than unchanged M0 links (median TF–IDF similarity 0.078 versus 0.175), so M1 should remain a sensitivity specification until manual validation is completed.

A procurement-profile domain audit is included as a second-stage evidence overlay. It does not create buyer identifiers and does not add links. It classifies profile domains as supporting, neutral, weak, ambiguous, or missing evidence for the M1 alias decisions. The

profile-audited variant removes zero links and is byte-for-byte identical in headline counts to M1: 847 accepted links, 47.6% blocking coverage, 26.8% linking rate, and Jaccard 1.000 versus M1.

Contents

1 Purpose and Position

The M1 branch exists because M0 deliberately avoids external enrichment. That makes M0 clean and auditable, but it also means that buyer identity depends only on whatever BOAMP provides in the notice: SIRET if present, SIREN if present, otherwise a normalized buyer name. In practice, many digital/ICT notices do not carry a usable identifier, and the same public buyer can appear under multiple name variants.

M1 is therefore a sensitivity analysis, not a replacement. Its role is:

1. preserve the official M0 design as the reference result;
2. change only buyer reconciliation and candidate blocking;
3. quantify how many candidate pairs and proxy-recurrence links are recovered when missing buyer identifiers are partially repaired;
4. make every enriched identity and alias decision traceable; and
5. separate useful sensitivity evidence from validated ground truth.

The current repository decision is:

> M0 balanced is the main result and official survival handoff. M1 buyer-SIREN enrichment is a sensitivity analysis. The event remains a constructed proxy recurrence, not a verified legal renewal.

2 Input Data and Lineage

2.1 M0 reference inputs

M1 reads the M0 outputs as fixed reference inputs. The source population and survival eligibility are not recomputed from scratch under a different scope. The principal M0 inputs are:

- `data/processed/boamp_m0_sources.csv`;
- `data/processed/boamp_m0_candidate_pairs.csv`;
- `data/processed/boamp_m0_links_balanced.csv`;
- `data/processed/boamp_survival_m0_balanced.csv`.

The M1 consistency audit records SHA-256 traces for these M0 files and checks that they remain read-only reference inputs. This is important: M1 does not silently rewrite M0.

2.2 SIREN enrichment dataset

The direct buyer enrichment source is the Hugging Face dataset `Data-Gouv-ML/jointure-boamp-siren-cote-acheteurs-2024-2025-et-2026`, commit `4bff9b1c5d2f4ad12834174e042828b7e52013d9`, downloaded on 2026-07-15. The join key is:

$$\texttt{B\_17\_idweb} \longrightarrow \texttt{notice\_id}.$$

The live audit found no duplicate notice identifiers in the three Parquet files, so the enrichment merge is one-to-one on notice ID for the years it covers.

Table 1: Notice-level SIREN enrichment files used by M1.

Year file	Rows	Date min	Date max	Overlap with M0 sources
2024	136,625	2024-01-01	2024-12-31	210
2025	137,423	2025-01-01	2025-12-31	189
2026	49,890	2026-01-01	2026-05-28	59
Total	323,938	–	–	458

Across the cleaned BOAMP table, 18,098 notices overlap the enrichment dataset. Among rows where BOAMP provides a SIRET-derived SIREN and the external enrichment also provides a valid SIREN, the agreement rate is 87.4% (6,937 agreements out of 7,939 rows). The 1,002 disagreements are not discarded silently; they are exported to `reports/tables/buyer_identifier_conflicts.csv` and flagged in the M1 identity provenance columns.

2.3 Procurement-profile dataset

The procurement-profile audit uses the data.gouv.fr dataset "Liste des profils d'acheteurs par entites adjudicatrices", downloaded as `data/raw/procurement_profiles/profils-acheteurs.csv` on 2026-07-15. The producer is DonnéesPubliques.org and the recorded license is `lov2`.

This dataset is evidence-only. Its row unit is a buyer-name / profile-domain / participation-domain summary row with a latest observed BOAMP date. It has no BOAMP notice ID, no SIREN, no SIRET, no city, and no postcode. Therefore M1 never uses procurement-profile domains as legal buyer identifiers and never creates a candidate pair from profile evidence.

Table 2: Procurement-profile dataset audit.

Metric	Value
Rows	23,260
Columns	5
Observed date range	2015-03-02 to 2020-12-31
Distinct normalized buyer names	15,039
Distinct normalized profile domains	4,700
Distinct participation domains	900
Malformed profile URL rows	396
Malformed participation URL rows	22
Exact duplicate buyer/profile/participation rows	1,715

3 End-to-End M1 Data Preprocessing

This section records the full preprocessing path from fixed M0 inputs to the M1-ready source table. It is intentionally written as a data lineage, not as an argument that M1 is superior to M0. The official M0 files remain fixed reference inputs throughout.

3.1 Step 0: freeze the M0 reference surface

M1 starts after the M0 preprocessing, candidate generation, linkage, and survival handoff already exist. The M1 script reads the M0 source table, candidate-pair table, balanced link table, and balanced survival handoff to recover the exact source population, score threshold, and comparison baseline. It does not recompute the BOAMP download scope, the digital/ICT filter, M0 duration imputation, or the official M0 balanced links. This design means every M1 difference can be interpreted as a buyer-identity sensitivity rather than a hidden change in the study population.

3.2 Step 1: load and audit the external SIREN files

`scripts/build_m1_buyer_siren_experiment.py` first loads the three Parquet files under `data/raw/buyer_siren_enrichment_m1/`. Only the documented BOAMP and Sirene columns listed in `ENRICHMENT_COLUMNS` are read. For each file the script records row counts, date coverage, missing notice IDs, duplicate notice IDs, missing SIREN/SIRET rates, file size, and SHA-256 hash. The audit is written to `reports/tables/buyer_siren_enrichment_audit.csv` and `buyer_siren_enrichment_audit.json`. The live audit supports the join key `B_17_idweb -> notice_id`: no duplicated notice IDs were found in the 2024, 2025, or 2026 enrichment files.

3.3 Step 2: merge notice-level enrichment without overwriting M0

The enrichment rows are renamed to explicit M1 fields and left-joined to `data/processed/boamp_clean_m0_no_enrichment.csv` by `notice_id`. The result is an all-notice enriched table, not yet the source-only analysis table. Dates are re-parsed after the merge so that subsequent temporal logic does not inherit object/string date types from the join. BOAMP buyer identifiers remain in their original fields; external SIREN/SIRET values are stored separately as enriched fields. This separation is central to the audit trail: M1 can use external evidence, but it does not rewrite BOAMP-provided SIRET/SIREN values.

3.4 Step 3: clean identifiers and classify provenance

The preprocessing then normalizes buyer names and cleans identifier fields. SIREN values must be nine digits; SIRET values must be fourteen digits; and validated external SIREN values must pass the identifier checksum before direct use. The script derives:

- BOAMP effective SIREN from BOAMP SIREN or from the first nine digits of a valid BOAMP SIRET;
- valid enriched SIREN and recovered enriched SIRET from the external notice-level join;
- buyer-name and department keys used only for alias auditing;
- disagreement flags where BOAMP SIRET-derived SIREN and enriched SIREN are both present but disagree.

Identifier conflicts are not resolved silently. They are exported to `reports/tables/buyer_identifier_conflicts.csv` and reflected in the identity-source and confidence columns.

3.5 Step 4: build the historical alias bridge

Direct enrichment only covers recent BOAMP notices. To test whether older name-only rows are fragmented, M1 builds a conservative historical bridge from directly enriched, non-conflicting rows. The bridge groups by normalized buyer name and department key. A historical alias is eligible for automatic propagation only when one valid enriched SIREN is uniquely observed for the name–department cell, the cell has no BOAMP conflict, the name is not a generic institutional label, and at least two notices support the mapping. Ambiguous cells are retained in `reports/tables/buyer_alias_ambiguous.csv` but are not propagated. The accepted bridge is written to `reports/tables/buyer_alias_bridge.csv`.

3.6 Step 5: construct M1 buyer identity fields

For every cleaned BOAMP notice, M1 applies the identity priority rule: preserve a BOAMP SIRET-derived identity first; otherwise use a valid direct enriched SIREN when there is no BOAMP conflict; otherwise use an automatic historical alias only for old rows with no BOAMP identifier; otherwise fall back to the original normalized buyer-name key. The output fields record the resulting M1 buyer key, the identity source, confidence label, direct versus inferred status, and conflict status. The source-only M1 table is then formed by selecting the same 3,159 source notices used by M0 and attaching those M1 identity/provenance fields. This table is written to `data/processed/boamp_clean_m1_buyer_enriched.csv`; the all-notice enriched table is written to `data/processed/boamp_clean_m1_buyer_enriched_all_notices.csv`.

3.7 Step 6: regenerate candidate pairs under the M1 buyer keys

Only after identity preprocessing is complete does M1 generate candidate pairs. The candidate universe keeps M0's source population, estimated end dates, temporal window, TF–IDF text representation, CPV hierarchy score, temporal score, composite weights, maximum of 30 candidates per source, and balanced threshold. The controlled change is buyer compatibility. A source and later candidate can enter the M1 candidate table through exact SIRET, same SIREN, historical alias, or original M0 name fallback. The resulting candidate table is `data/processed/boamp_m1_candidate_pairs.csv`.

3.8 Step 7: create links, survival rows, and diagnostics

M1 ranks candidates within each source by the M1 composite score, accepts the rank-1 candidate when its score is at least the M0 balanced threshold, and writes `data/processed/boamp_m1_links_balanced.csv`. The survival handoff keeps one row per eligible source and writes `data/processed/boamp_survival_m1_balanced.csv`. Downstream audit tables compare M0 and M1 at source and link level, identify incremental M1 links, summarize recovery mechanisms, measure candidate reuse, produce negative-control diagnostics, and export an unlabeled manual-review sample.

3.9 Step 8: profile-domain audit as a separate overlay

The procurement-profile step is not part of the core SIREN preprocessing. It reads M1 aliases and M1 candidate pairs after the buyer-enrichment branch has already been built. It normalizes profile domains, classifies them as buyer-specific, highly concentrated, shared institutional, generic platform, ambiguous, or malformed, and annotates M1 alias/candidate evidence. It may flag conflicts for manual review, but it does not create legal identifiers, does not create new buyer matches, and does not replace M0 or M1.

Table 3: End-to-end M1 preprocessing products.

Output	Role in the M1 preprocessing lineage
`buyer_siren_enrichment_audit.csv`	File-level audit of external SIREN/SIRET enrichment inputs.
`buyer_identifier_conflicts.csv`	Rows where BOAMP-derived and enriched legal identifiers disagree.
`buyer_alias_bridge.csv`	Automatically eligible historical name–department aliases.
`buyer_alias_ambiguous.csv`	Ambiguous or review-only aliases not propagated automatically.
`boamp_clean_m1_buyer_enriched_all_notices.csv`	All cleaned BOAMP notices after notice-level enrichment and provenance assignment.
`boamp_clean_m1_buyer_enriched.csv`	The same 3,159 eligible source notices as M0, with M1 identity fields attached.
`boamp_m1_candidate_pairs.csv`	M1 candidate pairs generated after buyer-identity preprocessing.
`boamp_m1_links_balanced.csv`	Accepted M1 proxy-recurrence links at the M0 balanced threshold.
`boamp_survival_m1_balanced.csv`	One-row-per-source M1 survival handoff with provenance columns.

4 Notation

For each eligible source or candidate notice i, define:

- d_i: publication date;
- $\hat{e}_i$: estimated end date inherited from M0;
- u_i: normalized buyer name;
- t_i: BOAMP SIRET, if format-valid and present;
- σ_i^B: BOAMP effective SIREN, using BOAMP SIREN if present or the first nine digits of a valid BOAMP SIRET;
- σ_i^E: valid enriched SIREN from the direct notice-level enrichment join;
- σ_i^A: SIREN inferred from an automatic name–department historical alias;
- k_i^0: original M0 buyer key;
- k_i^1: M1 primary buyer key;
- y_i: publication year.

Let H_i denote the historical-alias eligibility condition:

$$H_i = [y_i \leq 2023] \wedge \left[\sigma_i^A \neq \varnothing\right] \wedge [t_i = \varnothing] \wedge \left[\sigma_i^B = \varnothing\right].$$

So a historical alias can only fill a missing identifier for old notices. It does not overwrite a BOAMP SIRET or BOAMP SIREN.

5 M1 Buyer Identity Construction

5.1 Direct enrichment fields

The direct enrichment process first cleans the external SIREN/SIRET fields: non-digits are removed, SIREN must be nine digits, SIRET must be fourteen digits, and external SIREN must pass the checksum validation before it is considered valid for direct use.

The implementation preserves three separate ideas:

- BOAMP identifiers, which are the original source of truth for M0;
- enriched identifiers, which are external evidence joined by notice ID for 2024–2026 notices;
- conflict flags, which record disagreement between BOAMP SIRET-derived SIREN and enriched SIREN.

This separation is what prevents M1 from becoming an untraceable overwrite of M0.

5.2 Historical alias bridge

The historical alias bridge is built from directly enriched, non-conflicting notices. For each normalized buyer name and department key, M1 examines the set of observed valid enriched SIRENs. An alias is automatically propagated only if:

1. exactly one SIREN is observed for the name–department cell;
2. the cell has no BOAMP-SIRET conflict;
3. the buyer name is not a generic label such as `mairie`, `commune`, or `centre hospitalier`;
4. the support count is at least two notices.

The bridge contains 1,247 automatically eligible aliases and 1,912 ambiguous or review-only aliases. Ambiguous aliases are retained for audit but are not propagated.

5.3 Piecewise identity rule

The M1 SIREN value is constructed as:

$$\sigma_i^1 = \begin{cases} \sigma_i^B, & \text{if BOAMP effective SIREN exists,} \\ \sigma_i^E, & \text{else if valid enriched SIREN exists and no BOAMP conflict,} \\ \sigma_i^A, & \text{else if } H_i \text{ is true,} \\ \varnothing, & \text{otherwise.} \end{cases}$$

The M1 primary buyer key is then:

$$k_i^1 = \begin{cases} \texttt{SIRET:}t_i, & \text{if BOAMP SIRET } t_i \text{ exists,} \\ \texttt{SIREN:}\sigma_i^1, & \text{else if } \sigma_i^1 \text{ exists,} \\ \texttt{NAME:}u_i, & \text{else if normalized buyer name } u_i \text{ exists,} \\ \varnothing, & \text{otherwise.} \end{cases}$$

This rule is implemented in `scripts/build_m1_buyer_siren_experiment.py` in `apply_m1_identity`. The exact practical consequence is that M1 can move some M0 `NAME:...` rows into `SIREN:...` keys, but it does not replace an available BOAMP SIRET with an external value.

5.4 Confidence labels

The identity confidence labels are deliberately descriptive rather than probabilistic:

- `HIGH`: direct BOAMP SIRET, including conflict-flagged SIRET rows where BOAMP is still preserved;
- `MEDIUM_HIGH`: direct valid enriched SIREN joined by notice ID;

Table 4: M1 identity-source distribution among eligible sources.

Identity source	Source notices
NAME_FALLBACK	1,231
UNIQUE_NAME_DEPARTMENT_ALIAS	919
DIRECT_BOAMP_SIRET	686
DIRECT_ENRICHMENT_NOTICE_JOIN	296
DIRECT_BOAMP_SIRET_CONFLICT_FLAGGED	27

- MEDIUM: direct BOAMP SIREN without SIRET;
- MEDIUM_LOW: unique name–department historical alias;
- LOW: normalized-name fallback;
- CONFLICT: unresolved enriched identifier disagreement.

In the current eligible source population, the counts are 713 high, 296 medium-high, 919 medium-low, and 1,231 low.

6 Candidate Generation and Scoring

6.1 What stays identical to M0

M1 keeps the M0 design choices below:

- same eligible source population: 3,159 digital/ICT APPEL_OFFRE source notices;
- same estimated end date and duration-imputation logic;
- same TF–IDF vectorization for object text;
- same CPV hierarchy score;
- same temporal score centered on the source notice's estimated end date;
- same maximum of 30 candidates per source after temporal proximity ordering;
- same composite-score weights;
- same M0 balanced threshold: 0.323022.

The current M1 temporal window is six months. It is computed from the same median-duration rule used by the implementation and clipped to the configured floor/cap interval $[6, 24]$ months.

6.2 Buyer compatibility mechanisms

For a source notice i and later candidate notice j, M1 materializes a candidate only if at least one buyer-compatibility mechanism is satisfied:

$$
m_{ij} = \begin{cases} \texttt{EXACT\_SIRET}, & t_i = t_j \neq \varnothing, \\ \texttt{HISTORICAL\_ALIAS}, & \sigma_i^1 = \sigma_j^1 \neq \varnothing \text{ and either side uses an alias}, \\ \texttt{SAME\_SIREN}, & \sigma_i^1 = \sigma_j^1 \neq \varnothing, \\ \texttt{MO\_NAME}, & k_i^0 = k_j^0 \neq \varnothing, \\ \varnothing, & \text{otherwise.} \end{cases}
$$

If the same candidate is reachable through several mechanisms, M1 keeps the highest-priority mechanism in this order:

$$
\texttt{EXACT\_SIRET} > \texttt{SAME\_SIREN} > \texttt{HISTORICAL\_ALIAS} > \texttt{MO\_NAME}.
$$

The buyer-mechanism score is:

Table 5: Buyer mechanism scores used in the M1 composite score.

Mechanism	s_{buyer}
`EXACT_SIRET_RECONCILIATION`	1.00
`SAME_SIREN_RECONCILIATION`	0.90
`HISTORICAL_ALIAS_RECONCILIATION`	0.75
`MO_NAME_FALLBACK`	0.60

6.3 Composite score

For a candidate pair (i, j), M1 uses:

$$
S_{ij}^1 = 0.35\, s_{\text{text},ij} + 0.30\, s_{\text{cpv},ij} + 0.25\, s_{\text{time},ij} + 0.10\, s_{\text{buyer},ij}.
$$

The temporal component is:

$$
s_{\text{time},ij} = \max\left(1 - \frac{|d_j - \hat{e}_i|}{w},\, 0\right),
$$

where $w = 6$ months in the current run. Candidate j must be later than source i, and it must fall within the temporal window around the source's estimated end date. M1 then ranks candidates within each source by S_{ij}^1, keeps the top-ranked candidate, and accepts it if:

$$
S_{ij}^1 \geq 0.323022.
$$

7 M1 Outputs

The principal M1 output files are:

- `data/processed/boamp_clean_m1_buyer_enriched.csv`;
- `data/processed/boamp_m1_candidate_pairs.csv`;
- `data/processed/boamp_m1_links_balanced.csv`;
- `data/processed/boamp_survival_m1_balanced.csv`;
- `reports/tables/m0_m1_linkage_comparison.csv`;

- `reports/tables/m0_m1_link_status_changes.csv`;
- `reports/tables/m1_incremental_links.csv`;
- `reports/tables/m1_manual_validation_sample_unlabeled.csv`.

The survival handoff remains one row per eligible source. Compared with M0, it adds M1 provenance columns such as `buyer_key_m1_primary`, `buyer_identity_source`, `buyer_identity_confidence`, and `buyer_match_mechanism`.

8 M0 versus M1 Results

Table 6: Headline comparison between official M0 and M1 sensitivity.

Metric	M0 official	M1 sensitivity
Eligible source notices	3,159	3,159
Sources with at least one candidate	1,236	1,504
Zero-candidate sources	1,923	1,655
Blocking coverage	39.1%	47.6%
Candidate pairs	6,137	8,228
Accepted balanced links	618	847
Overall proxy-recurrence rate	19.6%	26.8%
Conditional acceptance among sources with candidates	50.0%	56.3%
Unique selected candidates	443	570
Candidate reuse rate	27.3%	29.5%
Maximum candidate multiplicity	6	8

The M1 increase is mostly a blocking effect. M1 recovers 269 sources that had no M0 candidate but receive at least one M1 candidate. At the final accepted-link level:

- 574 links are identical in M0 and M1;
- 273 accepted M1 links are not present in M0;
- 44 M0 links are not retained under M1 ranking/thresholding;
- 43 sources are linked in both methods but to a different selected candidate;
- link-set Jaccard overlap is 0.644.

8.1 Accepted links by mechanism

The historical-alias mechanism is the largest accepted-link source. This is exactly why M1 cannot be promoted to the official baseline without manual validation: the main incremental mechanism is useful, but weaker than direct BOAMP identifiers.

9 Incremental-Link Diagnostics

M1's incremental links should be read carefully. They are not random new events; they are links made possible by better buyer reconciliation. But their internal diagnostics are weaker than the unchanged M0 links.

Table 7: M1 accepted links by buyer-match mechanism.

Mechanism	Candidate pairs	Accepted links
`HISTORICAL_ALIAS_RECONCILIATION`	3,537	383
`MO_NAME_FALLBACK`	3,106	254
`EXACT_SIRET_RECONCILIATION`	883	120
`SAME_SIREN_RECONCILIATION`	702	90

Table 8: Incremental M1 links by recovery mechanism.

Recovery mechanism	Links	Median score	Median margin	Median text	CPV mismatch
Historical alias	198	0.360	0.066	0.073	25.8%
Same SIREN	75	0.388	0.073	0.085	14.7%

The key interpretation is:

$$\text{M1 improves candidate visibility} \not\Rightarrow \text{M1 proves extra legal renewals.}$$

The evidence is strongest when the buyer mechanism is exact SIRET or direct same-SIREN and weakest when the link depends on historical alias propagation plus low text similarity or low top1–top2 margin.

10 Procurement-Profile Domain Audit

10.1 Role in the pipeline

The profile-domain audit is a diagnostic overlay after M1. It answers: *when M1 says that two buyer-name variants belong together, do procurement-profile domains provide support, neutral shared-platform evidence, weak support, ambiguity, or no usable evidence?*

It does not:

- create a SIREN or SIRET;
- create a buyer key;
- create a candidate pair;
- increase the link count;
- replace M0 or M1.

10.2 Domain classification

URLs are normalized into domains and registered domains. Known shared platforms such as `achatpublic.com`, `marches-publics.gouv.fr`, `marches-securises.fr`, `boamp.fr`, and related platforms are classified as generic. Other domains are classified from concentration patterns:

- `GENERIC_PLATFORM`: broad multi-buyer platforms;
- `BUYER_SPECIFIC`: one buyer, enough repeated evidence;
- `HIGHLY_CONCENTRATED`: few buyers with one dominant buyer;

Table 9: Score diagnostics: unchanged M0 links versus incremental M1 links.

Median diagnostic	Unchanged M0 links in M1	Incremental M1 links
Composite score	0.407	0.372
TF–IDF text similarity	0.175	0.078
CPV score	0.100	0.100
Temporal score	0.825	0.836
Buyer mechanism score	0.750	0.750
Top1–top2 margin	0.080	0.070

- `SHARED_INSTITUTIONAL`: limited shared institutional use;
- `INSUFFICIENT_DATA`, `AMBIGUOUS`, or `MALFORMED`.

The audit found 1,408 generic-platform domains and 433 buyer-specific or highly concentrated domains.

10.3 Evidence status

Table 10: Profile evidence over M1 aliases and incremental links.

Profile evidence status	Alias rows	Incremental links
`NO_PROFILE_EVIDENCE`	2,423	76
`NEUTRAL_SHARED_PLATFORM`	593	59
`STRONG_SUPPORT`	110	86
`AMBIGUOUS`	29	2
`WEAK_SUPPORT`	4	50

For the 1,247 auto-propagation aliases specifically, the profile evidence counts are: 928 no profile evidence, 253 neutral shared platform, 53 strong support, 12 ambiguous, and 1 weak support. For the 198 historical-alias incremental links, the counts are: 68 strong support, 54 neutral shared platform, 41 weak support, 34 no profile evidence, and 1 ambiguous.

10.4 Profile-audited variant

The profile-audited variant only removes explicit profile conflicts among historical-alias candidate pairs. It does not add new pairs. In the current run it removes zero links:

11 Workflow Diagrams

12 Consistency Checks

The M1 consistency audit contains ten checks, all passing:

- no accidental many-to-many notice merge;
- no duplicated notice IDs in M1 sources;
- no silent overwrite of BOAMP SIRET;

Table 11: M1 versus profile-audited M1.

Metric	M1	Profile-audited M1
Eligible sources	3,159	3,159
Sources with candidates	1,504	1,504
Candidate pairs	8,228	8,228
Accepted links	847	847
Blocking coverage	47.6%	47.6%
Linking rate	26.8%	26.8%
Candidate reuse rate	29.5%	29.5%
Jaccard versus M1	1.000	1.000
Links removed versus M1	0	0

- every enriched identifier has provenance;
- all 1,002 conflict rows are traceable;
- every accepted M1 link exists in the candidate table;
- every event row has a valid candidate;
- censored rows do not contain accepted candidates;
- event and censoring times are positive;
- M0 reference files are present for hash trace.

The profile-audit consistency layer contains fourteen checks, also all passing. The most important are that profile domains are not used as legal identifiers, generic platforms do not create buyer matches, explicit conflicts are traceable, every accepted audited link exists in the candidate table, and M0/M1 outputs remain read-only reference inputs.

13 Limitations

M1 improves coverage, but it introduces additional assumptions. The main limitations are:

1. **External-enrichment coverage starts in 2024.** Direct notice-level enrichment does not cover the full 2015–2026 study period, so historical propagation is needed for older notices.
2. **Identifier conflict is real.** The audit finds 1,002 cases where BOAMP SIRET-derived SIREN and enriched SIREN disagree. Those cases are flagged, but they show why enrichment cannot be treated as automatically superior to BOAMP.
3. **Historical aliases are plausible, not legal proof.** The automatic alias bridge requires unique SIREN evidence by normalized name and department, but it is still a heuristic.
4. **Incremental links have weaker text evidence.** The median text similarity of incremental links is 0.078, much lower than the unchanged M0 links at 0.175.
5. **Profile-domain evidence is incomplete and name-based.** It has no notice identifiers and no legal identifiers, so it can audit or flag support/conflict but cannot identify buyers.
6. **The event is still a proxy recurrence.** M1 changes the constructed candidate set; it does not observe legal contract renewal status.

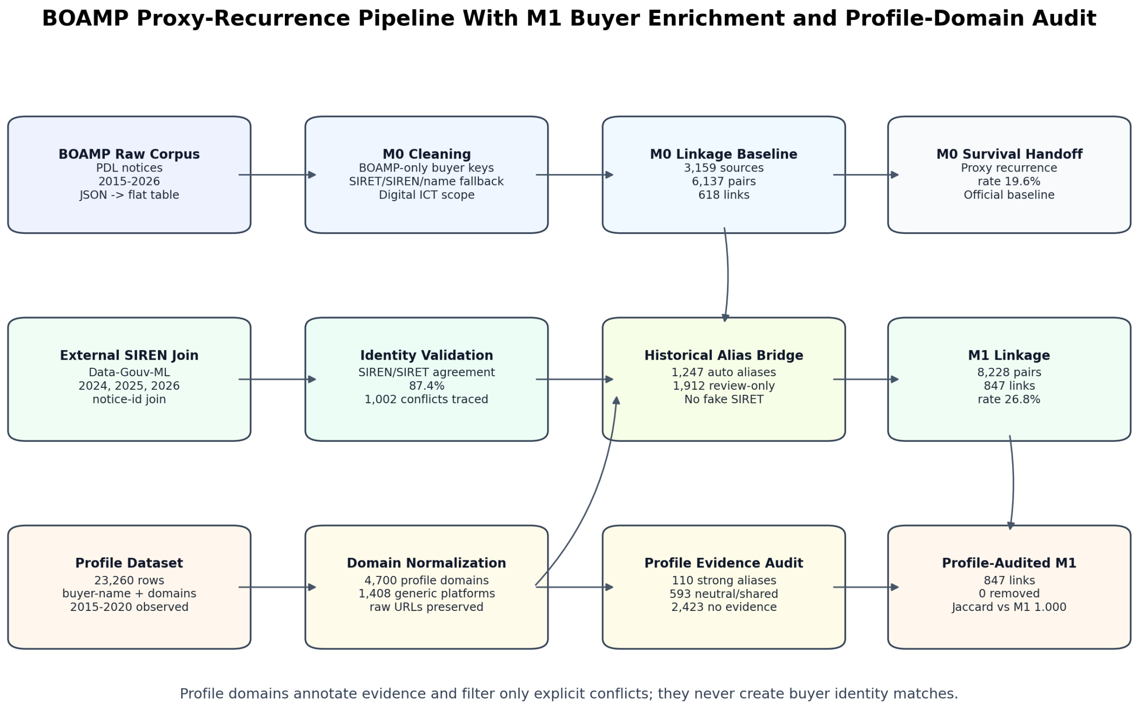

Figure 1: M1 enrichment and profile-domain audit workflow.

14 Recommended Interpretation

The correct reading is:

M0 is conservative and official;
M1 is wider and informative;
manual validation decides whether M1 can become primary.

M1 demonstrates that buyer-name fragmentation materially suppresses M0 candidate coverage. However, the additional links should be reported as a sensitivity result, not pooled with M0 as ground truth. The best immediate use of M1 is to guide manual validation: prioritize incremental links from historical aliases, low-margin links, CPV mismatches, and rows with profile ambiguity or missing profile evidence.

15 Reproducibility

The M1 enrichment and audit outputs are regenerated by:

```
python3 scripts/build_m1_buyer_siren_experiment.py
python3 scripts/build_m1_profile_domain_audit.py
MPLCONFIGDIR=/tmp/matplotlib-m1-report \
  python3 scripts/make_m1_academic_diagrams.py
```

The consolidated report itself is rebuilt from `reports/` with:

```
latexmk -cd -pdf -interaction=nonstopmode \
  -halt-on-error boamp_m1_technical_report.tex
```

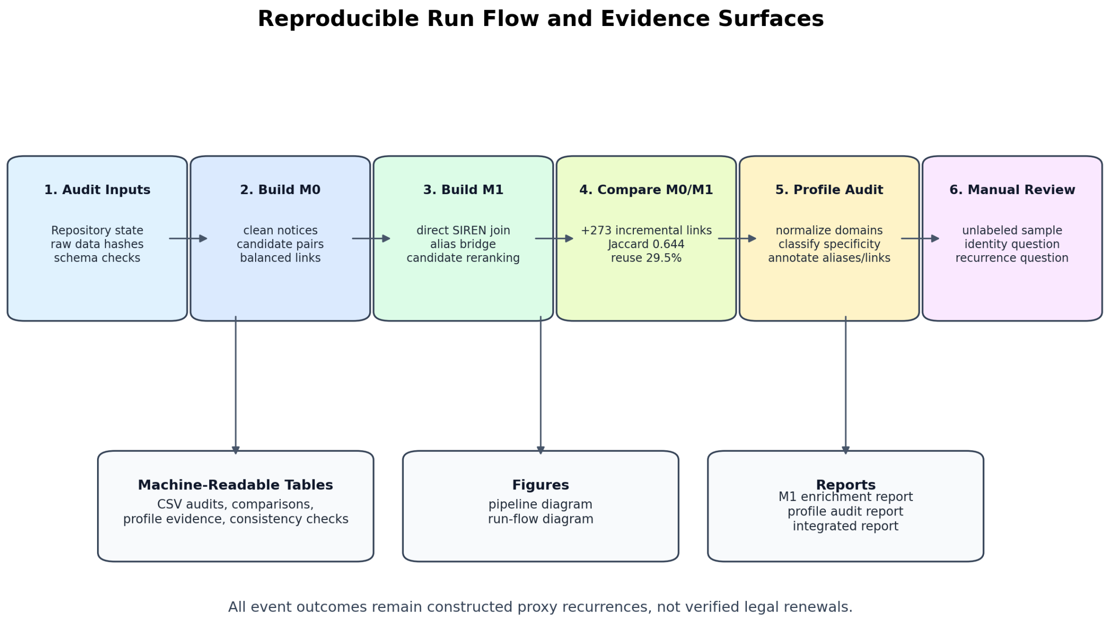

Figure 2: Reproducible run flow for the M1 experiment and audit outputs.

16 Relation to Previous M1 Notes

This document consolidates the previous M1 Markdown notes into one canonical report. The short generated notes separated the SIREN enrichment experiment, the procurement-profile audit, and the integrated diagram summary. This report keeps those parts together so the reader can follow the full chain:

$$
\begin{aligned}
\text{M0 reference} &\rightarrow \text{SIREN enrichment} \rightarrow \text{M1 candidate construction}\\
&\rightarrow \text{M0/M1 comparison} \rightarrow \text{profile-domain audit}\\
&\rightarrow \text{interpretation and limits.}
\end{aligned}
$$

17 Conclusion

M1 is technically useful and scientifically important because it shows that buyer-identity repair changes the observable recurrence signal. On the same eligible source population and the same threshold, it raises blocking coverage from 39.1% to 47.6% and accepted proxy-recurrence links from 618 to 847. That is a material sensitivity result.

The conservative conclusion is equally important: M1 should not replace M0 yet. The largest incremental channel is historical alias reconciliation, the incremental links have lower median text similarity than unchanged M0 links, and profile-domain data are audit evidence rather than legal identifiers. The current repository position is therefore methodologically consistent: report M0 as the official BOAMP-only baseline, report M1 as a buyer-enriched sensitivity analysis, and use the exported manual-validation samples to decide whether any future report can promote an enriched specification.